

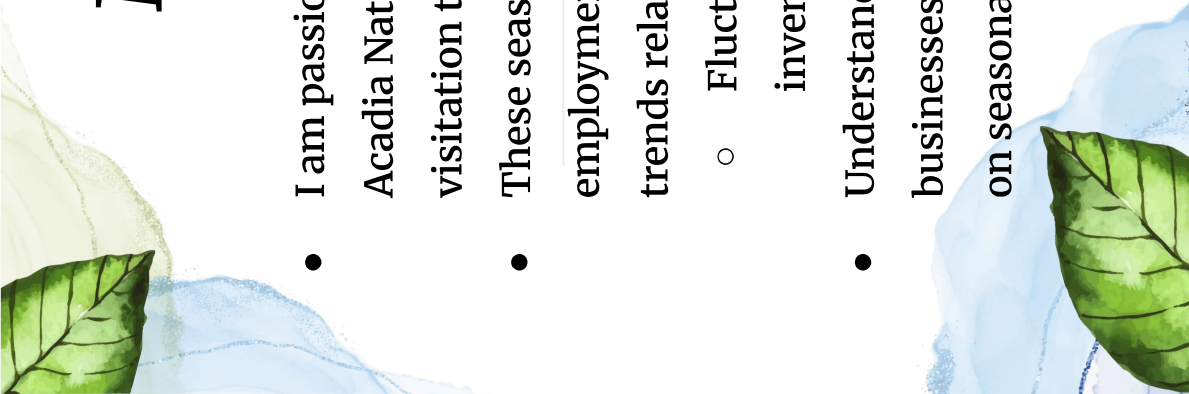
Tourism and Jobs in Maine: Evaluating Acadia National Park's Impact on Local Employment

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Motivation & Business Problem

- I am passionate about National Parks and the recreational opportunities they provide. Acadia National Park, the closest park to Boston, experiences strong seasonal visitation trends.
- These seasonal patterns create fluctuations in local economic activity, impacting employment and demand for local businesses, motivating an analysis of how visitor trends relate to employment.
 - Fluctuating demand makes it challenging for businesses to plan staffing, inventory, and operations efficiently.
- Understanding which industries are most sensitive to visitor trends can help businesses and policymakers allocate resources, anticipate labor needs, and capitalize on seasonal economic activity.



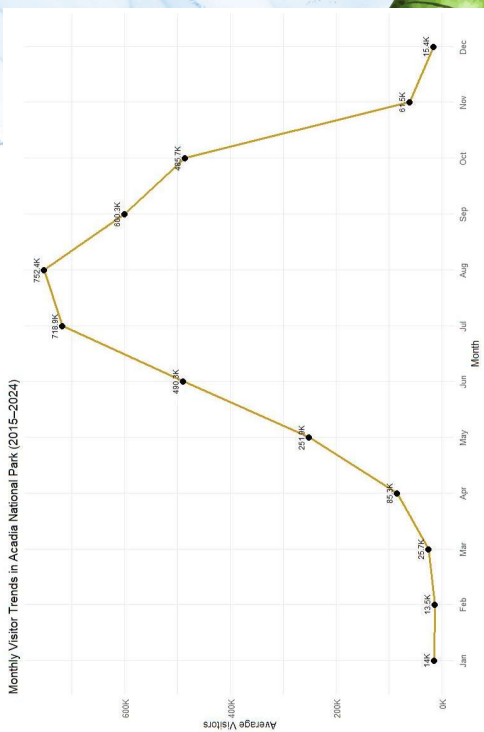
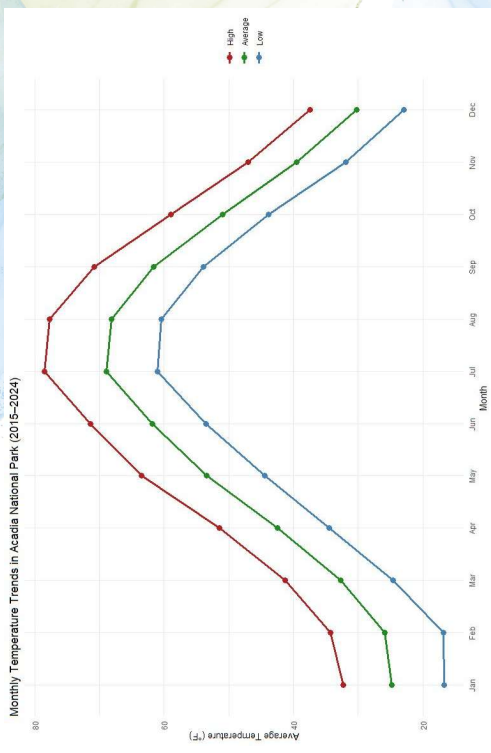
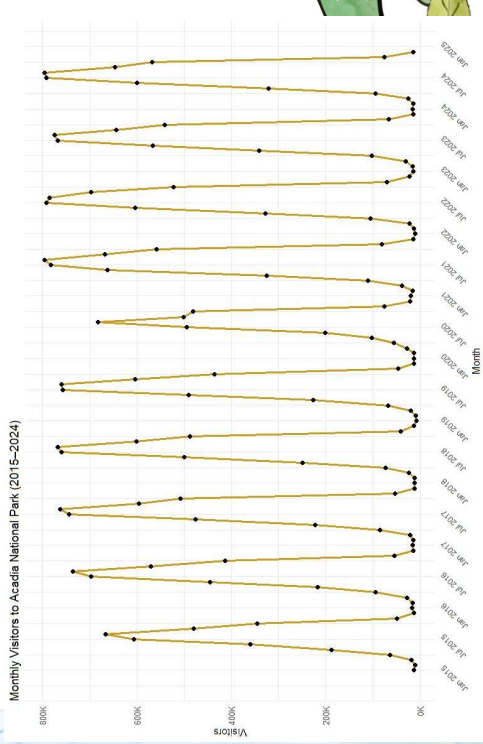
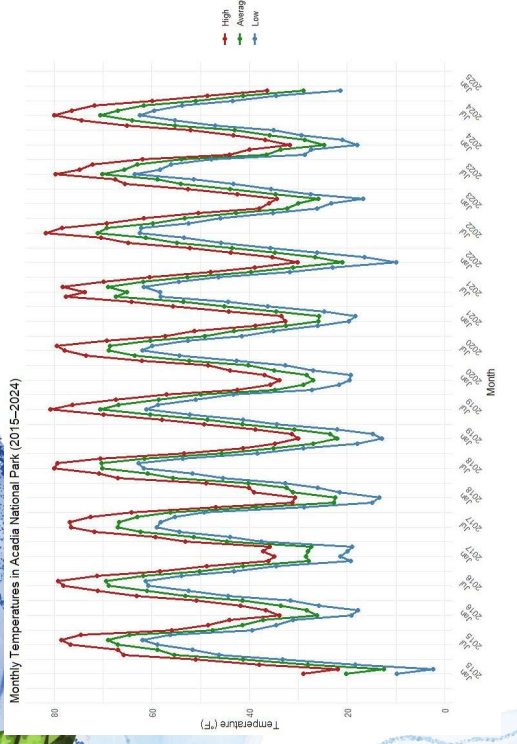


Data Sources & Key Methods

- Three necessary data sources: weather, visitors, and employment
 - [Weather data](#) from National Oceanic and Atmospheric Administration (NOAA)
 - [Acadia visitor data](#) from National Park Service (NPS)
 - [Employment data](#) from Bureau of Labor Statistics (BLS)
 - Primary Goals:
 - Analyze how weather patterns relate to visitor trends at Acadia National Park
 - Predict average monthly visitors from average monthly temperatures using both linear and logistic regression
 - Examine how visitor trends influence employment nearby
 - Predict average monthly employment by industry from average monthly visitors using linear regression, and determine which industries are the most seasonally volatile
 - All data used to compute monthly averages spans from January 2015 – December 2024 (10 years)
 - Focused on Hancock County (county containing Acadia NP) for employment analysis
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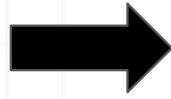
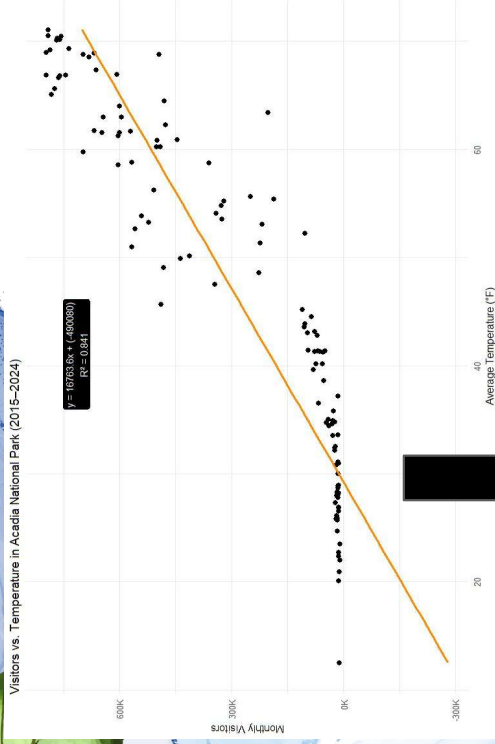
Results

Unsurprisingly, there is a clear seasonal temperature trend

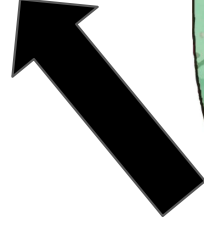
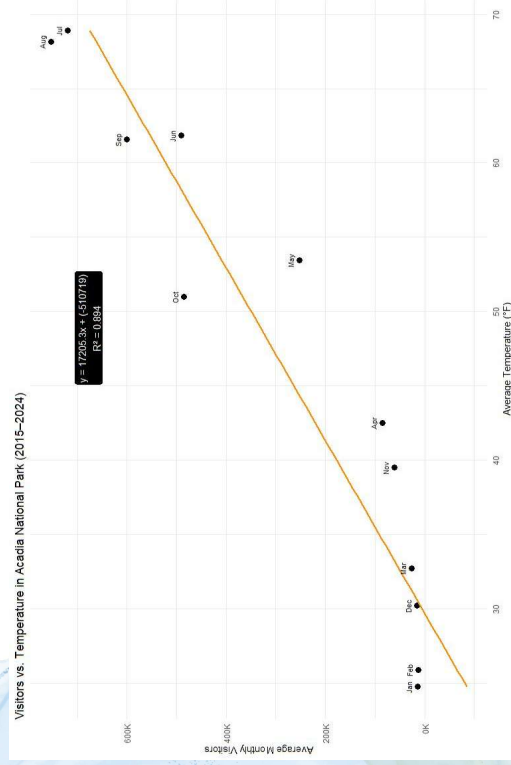
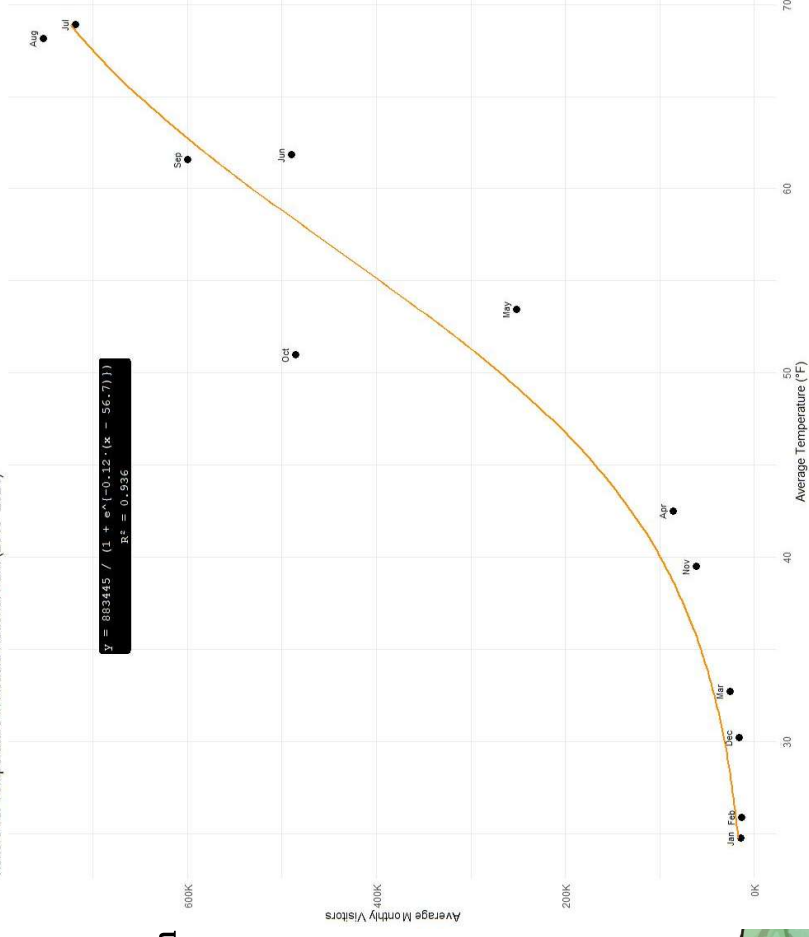


Monthly visitorship to the park follows a similar trend

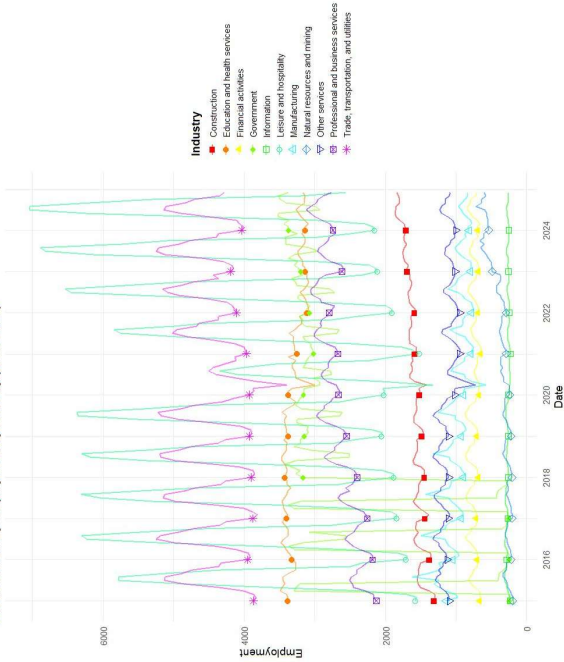
Results



Visitors vs. Temperature in Acadia National Park (2015–2024)



Hancock County Employment by Industry (2015-2024)



Results

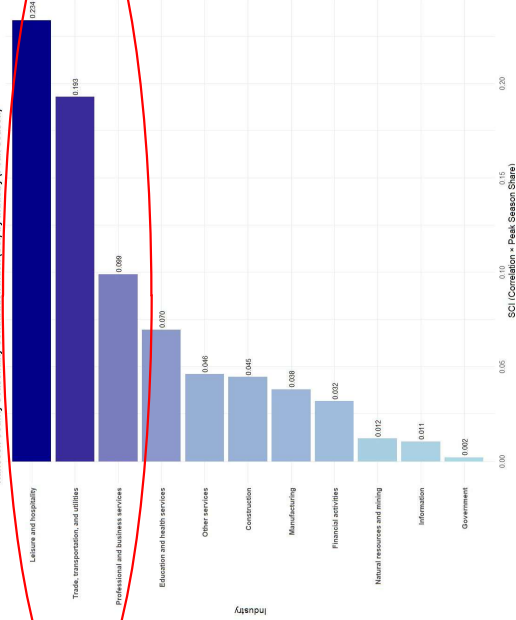
Determining the most seasonally volatile industries in Hancock County



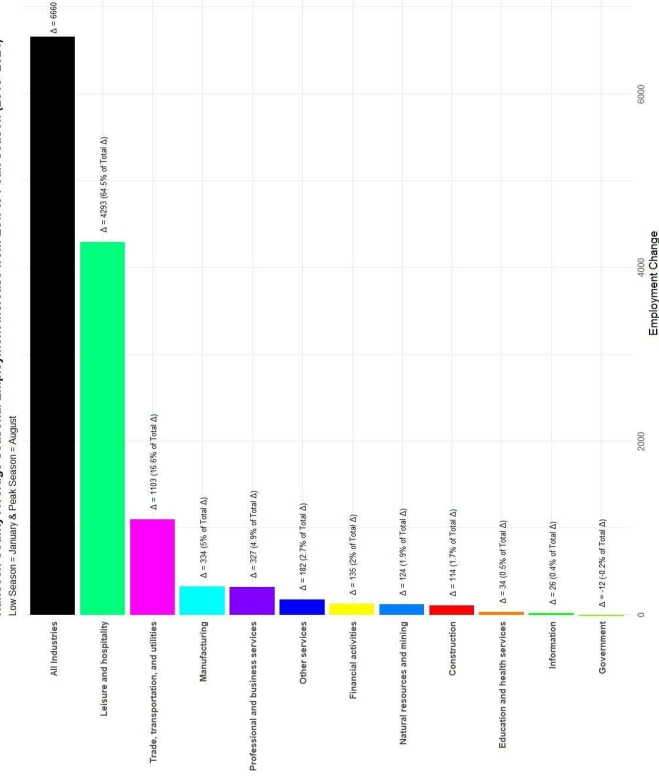
*Peak Season is defined as July to September here

Seasonality Contribution Index (SCI) is determined by correlation and park visitorship multiplied by peak season industry share

Hancock County Seasonality Contribution Index (SCI) by Industry (Peak Season)



Hancock County Average Seasonal Employment Increase from Low to Peak Season (2015-2024)



The industries that contributed most to the increase from lowest month (January) to highest (August) is very similar to the results of the SCI analysis

Results

Top 3 Industries: (For deeper analysis)

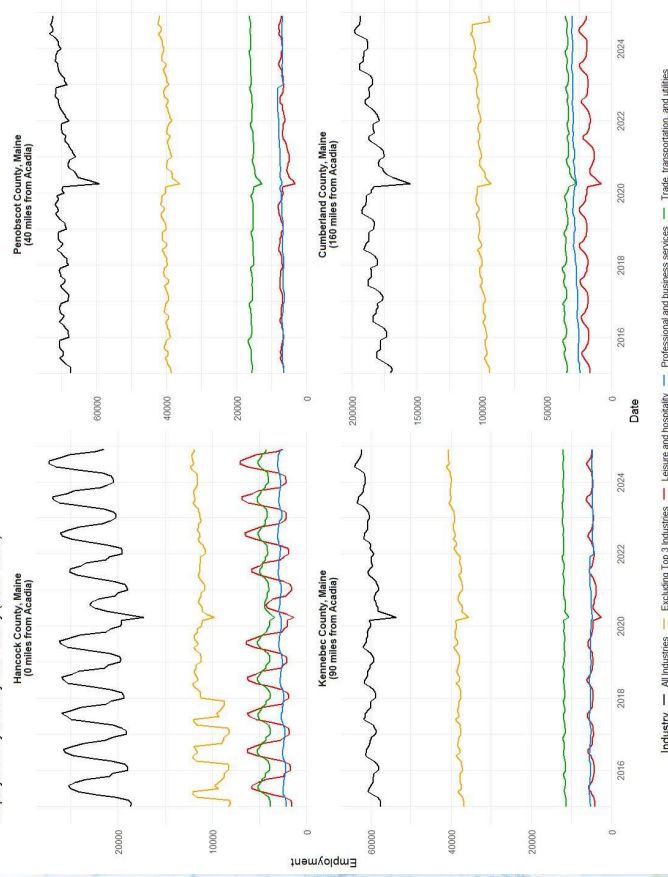
1. Leisure and hospitality
2. Trade, transportation, and utilities
3. Professional and business services

Hotels, campgrounds, restaurants, tour companies, etc.

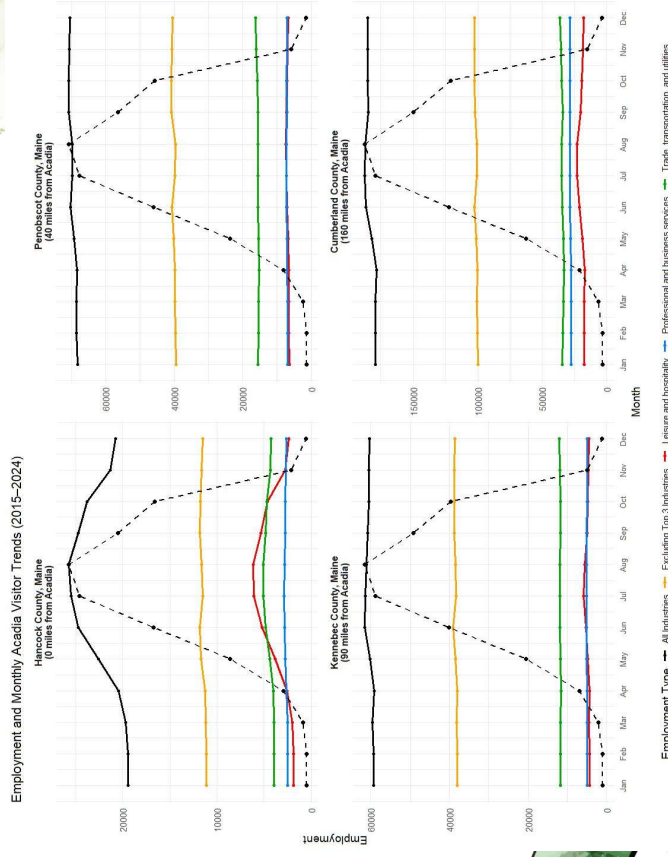
Local shops, gift stores, ferry operators, shuttle services, etc.

Accounting, marketing, or management services supporting local businesses

Employment by Industry and County (2015–2024)

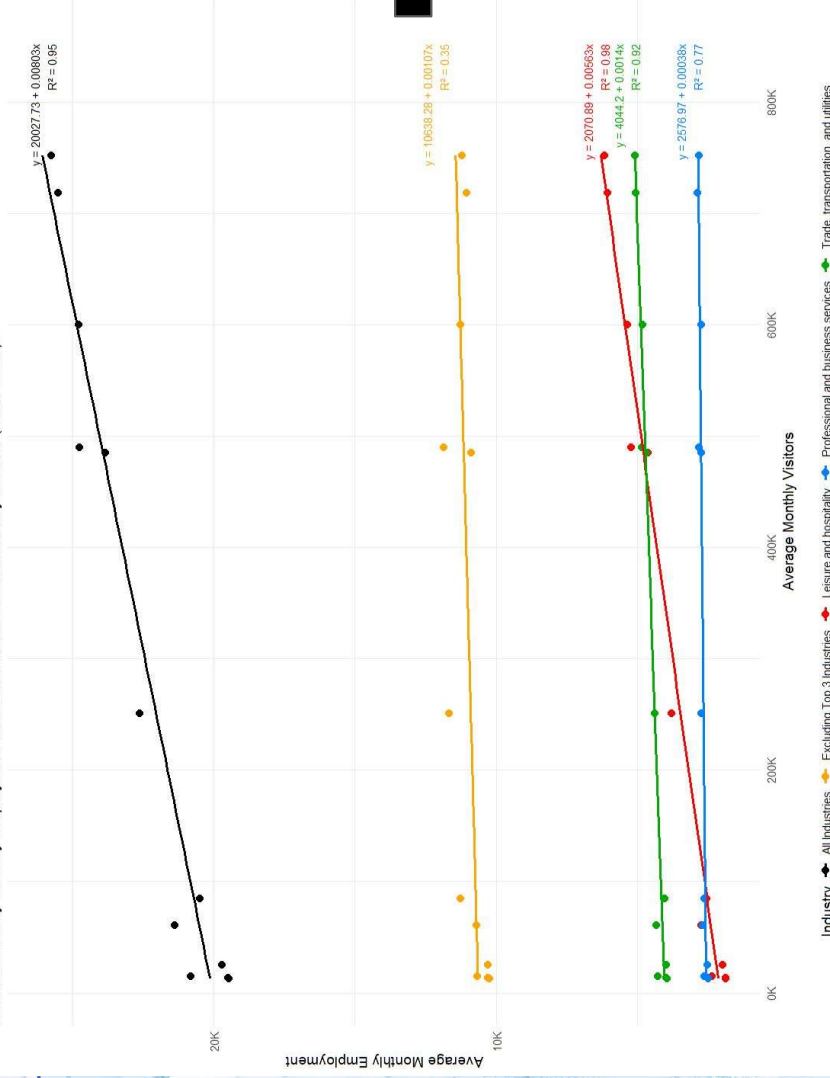


Evaluating how employment trends differ in other neighboring counties (control group)



Results

Hancock County Monthly Employment vs. Acadia National Park Monthly Visitors (2015–2024)



Modeling the relationship between visitorship and employment

Employment Sensitivity to Park Visitation
Slope and Inverse Slope of Employment ~ Visitors Regression

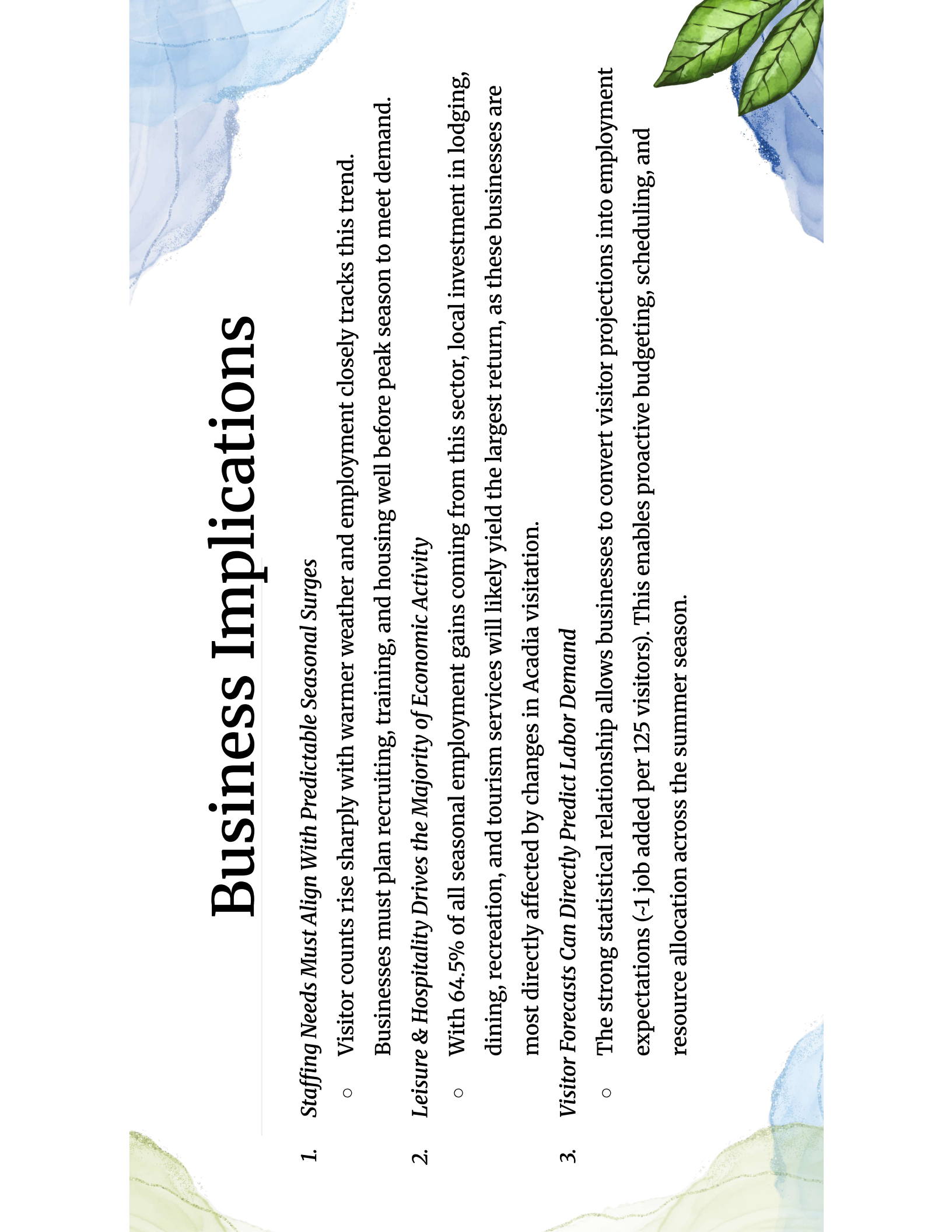
Industry	Slope	Inverse Slope
All Industries	0.00803	125
Excluding Top 3 Industries	0.00107	936
Leisure and hospitality	0.00563	178
Professional and business services	0.00038	2,638
Trade, transportation, and utilities	0.00140	712



For every additional X visitors, the employment in the respective industry increases by 1

Main Conclusions

- Visitor activity is highly seasonal and strongly influenced by weather, with significantly higher visitation during warmer months. This creates predictable peaks in local economic activity.
- Employment in Hancock County closely follows visitor trends, particularly in industries that rely on tourism. As visitation rises, overall employment increases in a measurable and consistent way.
- Leisure & Hospitality is the most sensitive industry to visitor volume, showing the strongest statistical relationship (highest slope and $R^2 = 0.98$). This confirms that Acadia National Park tourism is a primary driver of labor demand in this sector.
- Seasonal employment swings are substantial: from January to August, total employment rises by roughly 6,660 workers, representing more than a 30% increase.
 - Leisure & Hospitality accounts for the majority (64.5%) of this seasonal gain, highlighting its central role in the region's labor market. Elasticity of labor demand is high in tourism-dependent industries: approximately 1 job is added in Leisure & Hospitality for every 178 additional monthly visitors to Acadia.
- Impacts are geographically concentrated. Hancock County shows strong seasonal employment cycles tied to visitor levels, while more distant counties exhibit minimal or no fluctuation, indicating that the economic effects of park tourism are highly localized.



Business Implications

1. *Staffing Needs Must Align With Predictable Seasonal Surges*
 - Visitor counts rise sharply with warmer weather and employment closely tracks this trend.
Businesses must plan recruiting, training, and housing well before peak season to meet demand.
2. *Leisure & Hospitality Drives the Majority of Economic Activity*
 - With 64.5% of all seasonal employment gains coming from this sector, local investment in lodging, dining, recreation, and tourism services will likely yield the largest return, as these businesses are most directly affected by changes in Acadia visitation.
3. *Visitor Forecasts Can Directly Predict Labor Demand*
 - The strong statistical relationship allows businesses to convert visitor projections into employment expectations (~1 job added per 125 visitors). This enables proactive budgeting, scheduling, and resource allocation across the summer season.